
PAMF: Prior-Aware Multimodal Fusion for Incomplete Time Series Data

Ziwen Kan

Wugeng Zheng

Tianlong Chen²Song Wang¹¹Department of Computer Science, University of Central Florida²Department of Computer Science, University of North Carolina at Chapel Hill

Abstract

In healthcare, multimodal time series tasks often operate on incomplete observations in practice, for example when ECG segments are lost because electrodes detach or an entire respiratory channel is unavailable during overnight monitoring. Such missingness typically appears in two structurally distinct patterns: within-modality missing, where values are absent within an otherwise observed modality, and modality-level missing, where an entire modality is unavailable. Existing methods typically represent unobserved data implicitly through masks or missing embeddings, without learning instance-specific missing information, and most are designed for only one missingness pattern. A natural approach is to explicitly estimate the missing data; however, existing imputation methods treat missingness uniformly despite their different structural priors, and the imputation process is often isolated from downstream tasks, preventing downstream tasks from guiding imputation toward more informative representations. To address these limitations, we present PAMF, a multimodal time-series framework that explicitly handles different missingness patterns while coupling imputation with downstream prediction through prior-aware flow matching and weight sharing. Specifically, the method initializes the flow-matching source state with type-specific priors to distinguish two missing types. It further connects imputation and classification through architecturally matched encoders with weight sharing, transferring task-relevant representations into the imputation process. Experiments on multiple multimodal healthcare time-series benchmarks show that the proposed method achieves the strongest overall downstream performance across diverse datasets and missing settings compared with existing baselines.

1 Introduction

Multimodal time series are central to healthcare applications [53, 36, 38] and beyond [41, 13], where heterogeneous sensors jointly capture various information like different trends and periodic patterns. [22, 19]. However, in real-world deployment, these benefits are often limited by pervasive missing data. For example, in sleep staging from overnight polysomnography, it is often difficult to obtain complete physiological recordings for every subject-night. Missing data takes two structurally distinct forms: *within-modality missing*, where individual time points are absent within an otherwise recorded modality, for example when a sensor goes offline for a short period of time, and *modality-level missing*, where an entire modality goes unrecorded [21, 46, 11], for example when a patient is not in a condition to undergo a particular examination such as electrocardiogram (ECG) or respiratory signal. If not handled properly, both types of missingness can seriously degrade downstream performance. How to effectively handle both types of missingness therefore remains a critical open problem in multimodal time series analysis [46, 22, 45].

Early multimodal time series studies assume all modalities are fully observed and readily aligned [41, 53, 52, 31], limiting their applicability under real-world missingness. Recently, few methods that do account for missing data take a conservative approach: either substituting absent modalities with learned representations [28] or routing around missing inputs via gating and masks [11], or memory banks [51], without learning instance-specific representations for the missing content, and most such methods address only one of the two missing types, limiting their applicability in realistic deployment scenarios. To the best of our knowledge, existing methods do not simultaneously tackle both challenges. Table 1 summarizes how they fall short across the key capabilities required.

Table 1: Comparison of existing methods against PAMF. Methods include MulT [41], TFN [52], MAG [31], FuseMoE [11], FlexMoE [51], Maestro [28], MIRA [18], and representative single-modality imputation models such as CSDI [39] and SSSD [26]. **Imp.**: explicit reconstruction; **E2E**: imputation and fusion trained end-to-end.

Method	Within Missing	Modality Missing	Imp.	E2E
MulT, TFN, MAG	×	×	<i>N/A</i>	<i>N/A</i>
FuseMoE	×	✓	×	<i>N/A</i>
FlexMoe	×	✓	×	<i>N/A</i>
Maestro	×	✓	×	<i>N/A</i>
MIRA	✓	×	×	<i>N/A</i>
Single-Modality Imputation	✓	×	✓	×
PAMF (Ours)	✓	✓	✓	✓

A natural way to close this gap is to explicitly reconstruct missing data to address both missing types and generate instance-specific representation. Yet despite this need, no imputation solution is proposed, which we attribute to two independent technical challenges. First, existing diffusion-based generative imputation methods [39, 26, 48] start from the same domain-agnostic source distribution $\mathcal{N}(\mathbf{0}, \mathbf{I})$ for all missing cases, without distinguishing the different structures of within-modality missing and modality-level missing. As a result, the generative process may not be tailored to the specific missing type it must handle. Second, imputation and downstream prediction have largely been treated as independent, sequential tasks [45]: the imputation module is trained separately from the downstream task, so task-relevant representations learned later for fusion and downstream task cannot be used to guide imputation, leaving the imputation process largely task-agnostic. Consequently, the recovered representations may not be adapted to the needs of the downstream task.

To address these challenges, we propose PAMF, a unified framework for multimodal time series analysis under missingness. For the first challenge, PAMF features a newly designed flow-matching-based [20, 25] generative module with *prior-aware initialization*, so that within-modality missing and modality-level missing can be handled with different initial states that reflect their distinct structural information. For within-modality missing, the initialization is constructed from temporal neighbor and mean values to keep the starting point within a plausible local observation range, following the intuition of classical carry-forward imputation [29]. For modality-level missing, the initialization is derived from other observed data to provide a starting point that better matches the underlying distribution. This design also reduces inference overhead to fewer function evaluations than diffusion-based alternatives [39, 26] require. For the second challenge, we introduce a weight-sharing mechanism, through which imputation and the downstream task are connected by transferring representations learned for downstream prediction into the imputation process. The resulting imputed signals are then used for downstream prediction to improve downstream performance. Our main contributions are as follows:

- We propose PAMF, a unified framework for multimodal time series analysis under missingness, with *prior-aware initialization* in a flow-matching-based generative module, so that within-modality missing and modality-level missing can be handled with different initial states that reflect their distinct structural information.
- We introduce a *weight-sharing mechanism* that connects imputation with the downstream task by initializing the imputation branch from representations learned for downstream prediction, allowing task-relevant structure to guide the imputation process.
- We evaluate PAMF on four datasets under multiple missingness settings, including within-modality missing, modality-level missing, and additional mixed/high-missingness cases, demonstrating the strongest overall downstream performance.

2 Related Work

Multimodal Time Series Fusion. Multimodal time-series fusion methods typically assume complete modalities. Tensor fusion networks [52], multimodal transformers [41], and multimodal

adaptation gates [31] demonstrate strong gains on sentiment and affective benchmarks; in the health-care domain, MISTS [53] extends this line to irregularly sampled electronic health records via attention-based modality alignment. Although achieving promising results, the assumption of full observation limits their use in realistic clinical settings where missingness is ubiquitous [4, 5].

More recent studies tried to tackle this issue by using masks, routing, or learned placeholders to implicitly represent missing, in line with a broader missing-modality multimodal learning literature that focuses on representation robustness rather than explicit signal recovery [46, 44, 47, 33, 6]. For modality-level missing, FuseMoE [11] routes inputs through a mixture-of-experts gating mechanism preventing the model from attending to missing-modality positions in the cross-encoder.; Flex-MoE [51] substitutes absent modalities with a learnable embedding bank, providing fixed representations that do not condition on the specific observed context; Maestro [28] applies adaptive sparse attention over partially observed streams, yet similarly leaves missing signals unaware. For within-modality missing, MIRA [18] handles irregular sampling and missing values through channel-independent sparse mixture-of-experts modeling, but remains prediction-oriented rather than an explicit cross-modal signal imputer. Related multimodal physiological and cross-modal generation studies also emphasize robustness representation under incomplete observations [14, 24], but they do not target the explicit imputation of missing.

Time Series Imputation. General time-series imputation primarily targets within-modality missing within observed channels, as summarized in recent surveys [45]. Earlier approaches include neural imputers such as M-RNN [49], BRITS [2], SAITS [8], and MVI [1] propagate observations across missing positions through recurrent or attention-based mechanisms; generative models such as GP-VAE [10], which places a Gaussian process prior over latent trajectories, and GAN-based models [27], which learn the data distribution adversarially. Motivated by denoising diffusion models [12], recent probabilistic imputers model conditional distributions over missing values more directly and then achieved better imputation results: CSDI [39] performs conditional score generation by treating observed time steps as conditioning signals via two-channel attention; SSSD [26] further incorporates structured state space models for long-range dependency modeling to estimate missingness; and LSCD [9] conditions diffusion on frequency-domain structure for time-series imputation. Related probabilistic formulations include spatiotemporal diffusion in PriSTI [23], Schrödinger-bridge-based imputation [3], consistent diffusion imputation [56], and optimal-transport-based imputation [43]. These methods improve signal-level imputation within observed multivariate series, but do not directly address multimodal modality-level missing. Unlike diffusion-based generation, which simulates stochastic denoising dynamics, flow matching learns a velocity field parameterized by the transport variable τ that defines an ODE transporting samples from a source distribution to the data distribution.

Flow Matching for Imputation. Flow matching [20, 25] therefore offers a natural alternative for generative imputation. Compared with diffusion-based iterative denoising, it reduces inference step to as few as 10 to 30 ODE steps and leaves the initial state x_0 designable, making prior-aware initialization practical. Recent imputation-oriented extensions follow the same direction: CLWF [30] studies Lagrangian Wasserstein flows for imputation; MIRI [50] frames imputation as minimizing data-mask mutual information under a rectified-flow objective; and CFMI [37] studies conditional flow matching for general missing-data imputation. Although these methods differ in path design and conditioning objectives, they share the same practical premise: the source distribution can encode useful structure before transport rather than starting from fixed noise. This makes flow matching particularly attractive for imputation, where observed temporal context or distribution information can inform initialization. However, existing flow-matching imputation methods still focus on tabular or unimodal missing-value recovery; they do not encode type-specific priors for within-modality missing and modality-level missing or couple signal-level imputation with downstream classification.

3 Method

3.1 Problem Formulation

For a sample with M modalities, modality m provides a time series $\mathbf{x}^{(m)} \in \mathbb{R}^{T \times C_m}$, where T is the shared sequence length and C_m is the number of channels. For each modality m , let $\mathbf{R}^{(m)} \in \{0, 1\}^{T \times C_m}$ be the binary observation mask, with value 1 for observed entries and 0 for missing entries. Let Ω_w denote the set of within-modality missing positions, and let Ω_m denote the set of

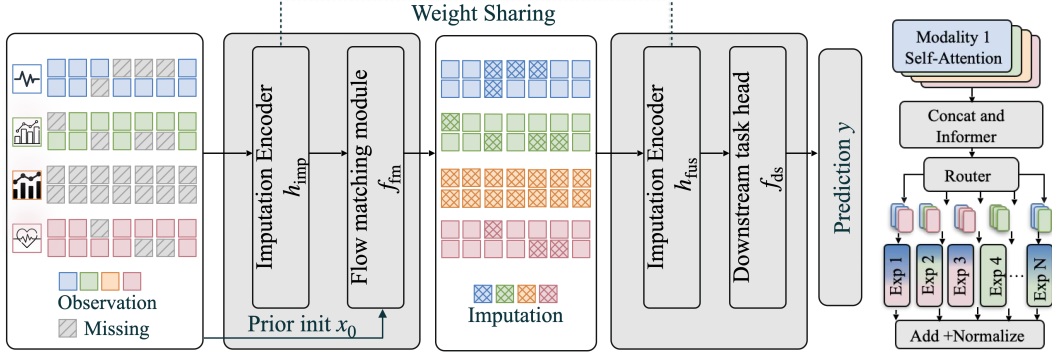


Figure 1: Overview of PAMF. Left: the end-to-end pipeline combines an imputation encoder h_{imp} , a fusion encoder h_{fus} , a flow matching module f_{fm} , and a downstream head f_{ds} . h_{imp} extracts the imputation context c_{imp} from the observed inputs. Conditioned on c_{imp} and a prior-aware source state x_0 , f_{fm} imputes the missing signals to generate completed data, which are then fed into h_{fus} to produce the fused representation c_{fus} for downstream head f_{ds} to predict y . Right: the shared MoE encoder backbone used by both encoders.

modality-level missing positions, where an entire modality is absent and thus $\mathbf{R}^{(m)} = \mathbf{0}_{T \times C_m}$. We write $\Omega = \Omega_w \cup \Omega_m$ for the full set of positions to impute. Given partially observed multimodal data and a downstream label y , PAMF initializes missing positions in Ω_w and Ω_m with type-specific priors, transports these source states toward completed signals via flow matching, and then learns task-relevant multimodal representations from the imputed data for downstream prediction.

3.2 Framework Overview

PAMF addresses imputation and downstream prediction in a single end-to-end framework built from four components: an imputation encoder h_{imp} , a fusion encoder h_{fus} , a flow matching module f_{fm} , and a task-specific downstream head f_{ds} . h_{imp} and h_{fus} share the same encoder structure. Concretely, each modality is first linearly projected and encoded by self-attention; the resulting modality embeddings are then concatenated and passed to a cross-modal MoE block, following sparsely gated expert routing [35] and prior multimodal studies [11], with Informer attention [55] to produce the fused context. Both encoders are designed to learn aligned multimodal representations and are coupled through three-phase staged training. In practice, h_{imp} provides the context representation c_{imp} used by the imputation branch, while h_{fus} provides the fused representation c_{fus} for downstream prediction. f_{fm} imputes missing time series values by transporting a structured source state x_0 to the target data distribution along straight ODE paths [20, 25], using a conditional velocity network driven by the observed signal and c_{imp} . Instead of fixing x_0 to a Gaussian distribution, we initialize it via *prior-aware initialization* that injects type-specific structural priors for within-modality missing and modality-level missing respectively (Section 3.3). f_{ds} is a task-specific downstream head applied to the fused representation c_{fus} to predict the final target y . Due to page limits, detailed implementation details are deferred to Appendix G.

3.3 Prior-Aware Flow Matching

The intuition for different prior is that starting state for imputation should depend on the missing type. For within-modality missing, nearby observed values in the same channel provide a local temporal prior. For modality-level missing, same-modality evidence is absent, so initialization may instead rely on dataset distribution information from the observation. Flow matching [20, 25] is well suited to encode this asymmetry because, unlike diffusion models with a fixed Gaussian start, it lets us design the source state x_0 while transporting it to the target signal x_1 through a continuous vector field. Here x_τ denote the source, target, and intermediate values on those positions. To let the generative path use information from the observed inputs, we condition the vector field on the imputation context c_{imp} produced by h_{imp} . Let $\Phi_\tau^\theta(\cdot; c_{imp})$ denote the resulting flow map induced by v_θ . The conditional

transport process can be written as

$$\begin{aligned} p_\theta(x_1 \mid \mathbf{c}_{\text{imp}}) &= [\Phi_1^\theta(\cdot; \mathbf{c}_{\text{imp}})]_\# p_0(x_0 \mid \mathbf{c}_{\text{imp}}, \Omega), \\ \frac{d\Phi_\tau^\theta(x_0; \mathbf{c}_{\text{imp}})}{d\tau} &= v_\theta(\Phi_\tau^\theta(x_0; \mathbf{c}_{\text{imp}}), \tau, \mathbf{c}_{\text{imp}}). \end{aligned} \quad (1)$$

Here $p_0(x_0 \mid \mathbf{c}_{\text{imp}}, \Omega)$ denotes the source initialization law, and $[\cdot]_\#$ is the push-forward operator. Equivalently, the state $x_\tau = \Phi_\tau^\theta(x_0; \mathbf{c}_{\text{imp}})$ follows the conditional ODE

$$\frac{dx_\tau}{d\tau} = v_\theta(x_\tau, \tau, \mathbf{c}_{\text{imp}}), \quad \tau \in [0, 1], \quad (2)$$

with the boundary condition $x_{\tau=0} = x_0$ and $x_{\tau=1} \approx x_1$. Here $\mathbf{c}_{\text{imp}}$ plays the standard role of conditioning the vector field at each FM step.

Following conditional flow matching [20, 40], we train this vector field by regressing to the conditional velocity along a chosen probability path. For the optimal-transport linear bridge used in rectified flows [20, 25], the path and target velocity u_τ are

$$x_\tau = (1 - \tau)x_0 + \tau x_1, \quad u_\tau(x_\tau \mid x_0, x_1, \mathbf{c}_{\text{imp}}) = x_1 - x_0, \quad \tau \sim \mathcal{U}[0, 1]. \quad (3)$$

This yields the masked conditional FM regression objective

$$\mathcal{L}_{\text{fm}} = \mathbb{E}_{\tau, x_0, x_1} \left[\frac{1}{|\Omega|} \sum_{i \in \Omega} \|v_\theta((1 - \tau)x_0[i] + \tau x_1[i], \tau, \mathbf{c}_{\text{imp}}) - (x_1[i] - x_0[i])\|_2^2 \right]. \quad (4)$$

Thus $\mathbf{c}_{\text{imp}}$ conditions the vector field at every FM step, while $\Omega = \Omega_w \cup \Omega_m$ restricts supervision to the missing entries.

Our main contribution is to condition the source state itself on the missing type. Generalized CFM does not require the source distribution to be Gaussian [40], which lets us replace an uninformative Gaussian source with a structured imputation prior. In practice, we construct x_0 directly with missing-type-specific rules. For each missing position $i = (t, c)$, we define

$$x_0[i] = \begin{cases} x_0^{\text{wm}}[i], & i \in \Omega_w, \\ x_0^{\text{ml}}[i], & i \in \Omega_m, \end{cases} \quad p_0(x_0 \mid \mathbf{c}_{\text{imp}}, \Omega) = p_0^{\text{wm}}(x_0^{\text{wm}} \mid \Omega_w) p_0^{\text{ml}}(x_0^{\text{ml}} \mid \mathbf{c}_{\text{imp}}, \Omega_m). \quad (5)$$

Thus, within-modality missing entries use a temporal-neighbor prior, whereas modality-level missing entries use a modality-level prior conditioned on $\mathbf{c}_{\text{imp}}$. The former exploits local continuity within the same channel, while the latter relies on information inferred from the present modalities. Both strategies reduce expected transport distance relative to a Gaussian start, consistent with the theoretical analysis in [25, 17]. In our implementation, the default within-modality missing prior is a rule-based temporal neighbor interpolation,

$$x_0^{\text{wm}}[t, c] = \alpha_{t,c} x_{t,c}^{\text{nbr}} + (1 - \alpha_{t,c}) \bar{x}_c^{\text{batch}}, \quad \alpha_{t,c} = \exp(-\lambda \cdot |t^* - t|), \quad (6)$$

where t^* denotes the time index of the nearest observed entry to t in channel c , so that $x_{t,c}^{\text{nbr}} = x_{t^*,c}$, and $\bar{x}_c^{\text{batch}}$ is the per-channel batch mean. Here $\lambda > 0$ is a fixed decay coefficient that controls how quickly the neighbor weight decreases as the temporal gap $|t^* - t|$ grows. We use this weighting because missing entries are more likely to follow their nearest temporal neighbor when the gap is short, whereas the global channel mean becomes a safer fallback as the gap widens. In the default implementation, we set $\lambda = 0.1$, yielding an exponential decay of neighbor confidence with temporal distance. The default modality-level missing prior is the per-channel batch mean:

$$x_0^{\text{ml}} = \bar{x}^{\text{batch}}. \quad (7)$$

This choice reflects the intuition that when an entire modality is absent, a conservative global statistic may provide a stable initialization that stays close to the typical value range of that channel before cross-modal conditioning refines it. We also evaluated richer variants, including a learned-neighbor weighting for x_0^{wm} that adapts the neighbor confidence, a batch-matching prior for x_0^{ml} that retrieves the most context-similar observed sample, and a memory-bank prior for x_0^{ml} that uses learnable cross-sample prototypes; Appendix C gives their full formulations, and Table 7 shows that they do not consistently improve over the simpler defaults above.

3.4 Weight Sharing and Three-Phase Training

We design h_{imp} and h_{fus} with the same encoder architecture. Both imputation and downstream prediction require temporal within-modality representation and cross-modal correlations, so the two branches could learn aligned rather than fully separate representations. Matched encoder structures therefore make $\mathbf{c}_{\text{imp}}$ and $\mathbf{c}_{\text{fus}}$ easier to align and support knowledge transfer across training stages. Because their learning objective and supervision signals differ, we couple them through a three-phase training strategy instead of a single joint loss.

Accordingly, the three phases use a downstream task loss and a flow matching velocity regression loss. Here $\mathbf{x}_{\text{obs}}$ is the incomplete input, and $\mathbf{x}_{\text{complete}}$ is the same input after FM imputation:

$$\mathcal{L}_{\text{ds}} = \text{CE}(f_{\text{ds}}(h_{\text{fus}}(\mathbf{x}_{\text{obs}})), y), \quad (8)$$

$$\mathcal{L}_{\text{fm}} = \mathbb{E}_{\tau \sim \mathcal{U}[0,1]} \left[\|v_{\theta}(x_{\tau}, \tau, h_{\text{imp}}(\mathbf{x}_{\text{obs}})) - (x_1 - x_0)\|_{\mathbf{M}}^2 \right], \quad (9)$$

where $x_{\tau} = (1 - \tau)x_0 + \tau x_1$ is the linear interpolation at FM time $\tau \in [0, 1]$ (distinct from the time-series index $t \in \{1, \dots, T\}$), $\mathbf{M} \in \{0, 1\}^{T \times C}$ is the binary imputation target mask (union of all missing positions; not the modality count M), x_1 is the ground-truth signal at those positions, and x_0 is the prior-aware source initialization from Section 3.3. $\mathcal{L}_{\text{ds}}$ is used in Phases 1 and 3 to learn and then refine predictive representations, whereas $\mathcal{L}_{\text{fm}}$ is used in Phase 2 to train the flow matching imputation module and reconstruct the observation from the prior initialization.

With these objectives in place, we train the framework in three phases.

Phase 1: Downstream task warmup. We train h_{fus} and f_{ds} with $\mathcal{L}_{\text{ds}}$ on the raw, partially observed multimodal input $\mathbf{x}_{\text{obs}}$, without any imputed augmentation. Starting from incomplete data forces the encoder to learn representations that remain informative under missingness, rather than relying on completed signals.

Phase 2: Flow matching training. h_{imp} is initialized by copying the weights of h_{fus} so that the imputation module starts from representations already shaped by the downstream task objective. The FM learns to impute conditioned on $\mathbf{c}_{\text{imp}}$, which encodes cross-modal context from the present modalities and provides the velocity field with information about the content of the missing signal.

Phase 3: Joint training on completed data. After Phase 2, h_{imp} and f_{fm} are frozen and used to impute all missing positions in producing the completed input $\mathbf{x}_{\text{complete}}$. We then train h_{fus} and f_{ds} with $\mathcal{L}_{\text{ds}}$ on $\mathbf{x}_{\text{complete}}$ to get final result.

Through this mechanism, weight sharing lets the FM branch inherit task-relevant encoder parameters, while imputed-data fine-tuning allows the recovered signals to improve the final downstream model; the effectiveness of this strategy is experimented and shown in Figure 3.

4 Experiments

4.1 Experimental Setup

Datasets. We evaluate on four physiological benchmarks. **Sleep-EDF** [16] is a polysomnography dataset for sleep staging, which we represent with four modalities: EEG (2 channels), EOG (1 channel), Resp (1 channel), and EMG (1 channel), and use for five-class sleep stage classification. **PTB-XL** [42] is a 12-lead ECG benchmark, which we partition into three lead-group modalities, Limb (I/II/III), Augmented (aVR/aVL/aVF), and Precordial (V1–V6), for five-class cardiac diagnostic classification (NORM/MI/HYP/CD/STTC). **PPG-DaLiA** [32] is a wearable sensing dataset originally released for heart-rate estimation; in our benchmark-adapted setting, we use five sensor modalities, chest ACC, wrist ACC, wrist BVP, wrist EDA, and wrist TEMP, for nine-class activity recognition. **Chapman-Shaoxing** [54] is a 12-lead arrhythmia corpus, which we represent as twelve single-lead modalities and use for 7-label diagnosis prediction. Detailed dataset statistics, split protocols, and preprocessing are provided in Appendix D.

Missing data protocol. We evaluate missingness under both basic and stress-test protocols. As basic settings, we consider within-modality missing and modality-level missing separately at a 20% missing rate. For within-modality missing, instead of adding Gaussian noise, we apply contiguous block masks with lengths drawn uniformly from [5%, 10%] of T , per channel, to mimic missingness

Table 2: Classification performance on four benchmarks under 20% missingness, with within-modality missing and modality-level missing evaluated separately. Means and standard deviations over three runs are rounded to three decimals. **Bold**: best; underlined: second-best.

Method	Sleep-EDF		PTB-XL		PPG-DaLiA		Chapman	
	F1	AUROC	F1	AUROC	F1	AUROC	F1	AUROC
<i>within-modality missing (20%)</i>								
FuseMoE [11]	0.552 $\pm$ 0.007	0.912 $\pm$ 0.006	0.243 $\pm$ 0.020	0.683 $\pm$ 0.011	0.602 $\pm$ 0.014	0.918 $\pm$ 0.008	0.311 $\pm$ 0.023	0.630 $\pm$ 0.007
Flex-MoE [51]	<u>0.668</u> $\pm$ 0.007	<u>0.967</u> $\pm$ 0.003	0.422 $\pm$ 0.022	0.731 $\pm$ 0.009	0.591 $\pm$ 0.005	0.917 $\pm$ 0.004	0.343 $\pm$ 0.003	0.708 $\pm$ 0.008
Maestro [28]	0.627 $\pm$ 0.003	0.955 $\pm$ 0.000	<u>0.622</u> $\pm$ 0.007	<u>0.876</u> $\pm$ 0.001	<u>0.692</u> $\pm$ 0.017	<u>0.943</u> $\pm$ 0.001	0.351 $\pm$ 0.024	0.762 $\pm$ 0.009
MIRA [18]	0.582 $\pm$ 0.013	0.936 $\pm$ 0.010	0.531 $\pm$ 0.004	0.839 $\pm$ 0.008	0.672 $\pm$ 0.002	0.936 $\pm$ 0.002	<u>0.403</u> $\pm$ 0.013	<u>0.815</u> $\pm$ 0.017
PAMF (ours)	0.697 $\pm$ 0.023	0.977 $\pm$ 0.001	0.661 $\pm$ 0.003	0.884 $\pm$ 0.003	0.714 $\pm$ 0.017	0.948 $\pm$ 0.006	0.518 $\pm$ 0.037	0.870 $\pm$ 0.007
<i>modality-level missing (20%)</i>								
FuseMoE [11]	0.502 $\pm$ 0.005	0.890 $\pm$ 0.001	0.314 $\pm$ 0.019	0.662 $\pm$ 0.004	0.562 $\pm$ 0.005	0.901 $\pm$ 0.004	0.327 $\pm$ 0.003	0.644 $\pm$ 0.002
Flex-MoE [51]	<u>0.651</u> $\pm$ 0.005	<u>0.961</u> $\pm$ 0.001	0.447 $\pm$ 0.029	0.728 $\pm$ 0.009	0.570 $\pm$ 0.007	0.909 $\pm$ 0.000	0.357 $\pm$ 0.010	0.699 $\pm$ 0.006
Maestro [28]	0.646 $\pm$ 0.012	0.953 $\pm$ 0.003	<u>0.604</u> $\pm$ 0.010	<u>0.872</u> $\pm$ 0.002	<u>0.675</u> $\pm$ 0.013	0.931 $\pm$ 0.003	0.413 $\pm$ 0.015	0.758 $\pm$ 0.003
MIRA [18]	0.561 $\pm$ 0.003	0.925 $\pm$ 0.005	0.504 $\pm$ 0.007	0.826 $\pm$ 0.001	0.663 $\pm$ 0.003	<u>0.932</u> $\pm$ 0.003	<u>0.415</u> $\pm$ 0.024	0.827 $\pm$ 0.010
PAMF (ours)	0.653 $\pm$ 0.009	0.963 $\pm$ 0.005	0.648 $\pm$ 0.002	0.878 $\pm$ 0.002	0.691 $\pm$ 0.015	0.944 $\pm$ 0.005	0.492 $\pm$ 0.012	<u>0.823</u> $\pm$ 0.006

from transient sensor dropout or signal failure. For modality-level missing, we remove whole modalities to mimic missing from no recoding, for example, subjects who did not wear or record a given sensor. Beyond these basic settings, we further evaluate a mixed protocol where within-modality missing and modality-level missing occur simultaneously to simulate a more realistic clinical setting, for example, in ICU a patient may lack EEG record entirely while recorded vital signs are also temporally interrupted [15], as well as high-missingness protocols that increase the missing rate to test robustness under more severe observation loss.

Baselines and Metrics. We compare against four strong multimodal methods that handle missing modalities within two years. FuseMoE [11] routes inputs through modality-specific experts, tolerating absent streams via sparse gating. Flex-MoE [51] substitutes missing modalities with learned embeddings in a mixture-of-experts architecture. Maestro [28] applies adaptive sparse attention over partially observed modality streams. MIRA [18] is a channel-independent sparse MoE Transformer trained for multimodal classification under missingness. None of these methods explicitly distinguishes between within-modality missing and modality-level missing, nor do they reconstruct missing signal values. We report AUROC and macro-averaged F1 score for classification, averaged over available independent runs with 3 random seeds.

4.2 Comparison with Baselines

Basic missingness. Table 2 reports classification performance under the two basic 20% missingness settings. Since all four benchmarks involve skewed label distributions across sleep stages, diagnosis classes, or activity labels, and precision-recall-based evaluation is often more informative than ROC-based evaluation on imbalanced classification problems [7, 34], we treat macro-F1 as the primary metric in this clinical-style setting. Three observations are most important. (1) PAMF delivers the strongest overall classification performance, achieving the best F1 on all eight dataset-setting pairs and the best AUROC on seven of them, which supports the advantage of explicit imputation over only tolerating missing inputs. (2) The strongest baseline differs across datasets, yet our method still performs best overall: Flex-MoE is especially competitive on Sleep-EDF, particularly under modality-level missing, suggesting that its subset-aware routing and missing-modality bank are effective when whole modalities disappear; Maestro’s modality-aware encoding helps it to become the strongest baseline on PTB-XL and PPG-DaLiA; and MIRA is strongest on Chapman, suggesting that its continuous-time sparse modeling is helpful on this more challenging ECG benchmark. Even so, PAMF remains best overall, indicating that explicit imputation provides a more consistent route to downstream accuracy. (3) The clearest advantage appears in macro-F1, which is especially relevant here because minority classes matter in all four benchmarks and F1 is often more informative than AUROC for imbalanced clinical-style classification [7, 34].

Mixed and high missingness. Beyond the two basic 20% settings, Table 3 evaluates harder but realistic scenarios. The mixed setting combines within-modality missing and modality-level missing in the same sample, which better reflects clinical data. We also consider a more extreme 50%

Table 3: PTB-XL classification under mixed and high missingness. Mixed denotes simultaneous within-modality missing+modality-level missing at 20% each. High-rate settings evaluate either within-modality missing or modality-level missing at 50%. Means and standard deviations over three runs are rounded to three decimals. **Bold**: best; underlined: second-best.

Method	Mixed		within-modality missing 50%		modality-level missing 50%	
	F1	AUROC	F1	AUROC	F1	AUROC
FuseMoE [11]	0.211 $\pm$ 0.040	0.601 $\pm$ 0.003	0.176 $\pm$ 0.041	0.612 $\pm$ 0.002	0.237 $\pm$ 0.014	0.617 $\pm$ 0.021
Flex-MoE [51]	0.428 $\pm$ 0.024	0.757 $\pm$ 0.003	0.411 $\pm$ 0.008	0.708 $\pm$ 0.006	0.460 $\pm$ 0.005	0.709 $\pm$ 0.006
Maestro [28]	<u>0.608</u> $\pm$ 0.009	0.867 $\pm$ 0.002	<u>0.514</u> $\pm$ 0.012	<u>0.856</u> $\pm$ 0.003	<u>0.566</u> $\pm$ 0.030	<u>0.848</u> $\pm$ 0.004
MIRA [18]	0.503 $\pm$ 0.005	0.828 $\pm$ 0.003	0.455 $\pm$ 0.004	0.812 $\pm$ 0.001	0.505 $\pm$ 0.001	0.823 $\pm$ 0.005
PAMF (ours)	0.641 $\pm$ 0.011	0.867 $\pm$ 0.003	0.626 $\pm$ 0.011	0.865 $\pm$ 0.003	0.600 $\pm$ 0.012	0.852 $\pm$ 0.005

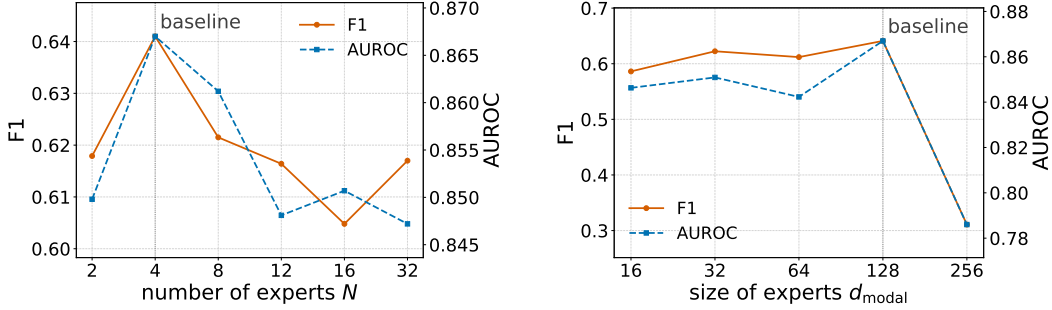


Figure 2: Hyperparameter sensitivity of our encoder configuration on PTB-XL under mixed 20% missingness. Left: sweep over the number of experts N . Right: sweep over the expert size d_{modal} .

missingness setting to test robustness under severe observation loss. As Table 3 shows, PAMF maintains the strongest overall downstream performance across all three settings. Under mixed missingness, it achieves the best F1 and matches the best AUROC, showing that its advantage persists when the two missingness mechanisms co-occur. Under 50% missingness, it continues to rank first on both settings. The clearest contrast is with Maestro under within-modality missing, where F1 falls from 0.622 in the 20% basic setting to 0.514 at 50%, a drop of 0.108. It is not surprising that while Maestro is strong with modality-level missing awareness, it does not specifically model within-modality gaps, so performance degrades more sharply when within-modality missing becomes heavy. Compared with Maestro, our imputation strategy is more resilient under heavy missingness.

4.3 Ablation Study

Hyperparameter. Our encoder introduces two key capacity controls: the number of experts N and the expert size d_{modal} . The former determines how finely the model partitions modality-specific patterns, while the latter controls the representation capacity of each expert. Both are important for performance, but both can also become harmful when enlarged without control. If N is too large, expert specialization becomes overly fragmented, making routing less stable and reducing effective data per expert. If d_{modal} is too large, the expert MLPs become more prone to overfitting under the limited supervision strategy. We therefore conduct the hyperparameter sweep in Figure 2 under mixed 20% missingness. As shown in the left panel, performance is strongest around $N=4$; after that point, the results begin to oscillate and decrease rather than improve, indicating that adding more experts mainly increases fragmentation rather than useful specialization. In the right panel, performance increases gradually before reaching its best point at $d_{\text{modal}}=128$, after which it drops sharply. Based on this trend, we use $N=4$ and $d_{\text{modal}}=128$ in the final configuration.

Figure 3 summarizes the effect of removing each major design component under mixed 20% missingness. All variants underperform the full model, confirming that the final performance relies on the cooperation of imputation, prior-aware source design, conditional cross-modal guidance, and encoder-level knowledge transfer. Among them, “No Imputation” yields the worst result, showing that directly optimizing the classifier on raw incomplete inputs is insufficient in the missing setting. Removing prior-aware initialization also leads to a clear performance drop and larger fluctuations,

Table 4: Comparison with diffusion-based imputation methods on PTB-XL under mixed 20% missingness (batch size = 64, GPU: B200). Each method is paired with the same downstream task component to isolate the generative imputation module. Performance is reported as Macro-F1. NFE denotes the number of function evaluations, and latency is wall-clock time per batch in milliseconds.

Method	Macro-F1	NFE	Latency (ms/batch)
CSDI [39]	0.626 ± 0.016	50	137.77
SSSD [26]	0.616 ± 0.009	50	526.98
PAMF (ours)	0.641 ± 0.011	20	75.07

suggesting that a structured source initialization is important for stabilizing the conditional generation process. Similarly, “No Context” degrades performance, indicating that conditioning the generative pathway on encoder-side context is also beneficial. “No Weight-Share” produces consistently lower results. We believe that without weight sharing, the imputation branch no longer receives sufficient guidance from the shared encoder representation shaped by the downstream classification objective. The “Prior Only” control performs better than “No Imputation” but still remains below the full model, indicating that the complete pipeline benefits from more than the prior signal alone.

4.4 Comparison with Diffusion Imputation

Strategy Effectiveness. We compare PAMF against diffusion-based imputation methods to assess whether flow matching provides a stronger generative imputation module under the same downstream task setting. For a fair comparison, we pair each diffusion baseline with the same downstream task component used in our framework, so the comparison isolates the generative imputation module rather than conflating it with differences in the classifier. Specifically, we evaluate the performance of CSDI [39] and SSSD [26] on PTB-XL under mixed 20% missingness. Table 4 shows that PAMF achieves the highest Macro-F1 while also delivering the lowest inference latency. This speedup comes from the FM transport, which reaches x_1 from x_0 with fewer function evaluations than iterative diffusion denoising. The performance gain is also consistent with our design: The source stage x_0 is initialized by the prior-aware rule in Equation (5), so the transport starts from a more informative point than the generic diffusion initialization used by CSDI and SSSD. Moreover, because these baselines are primarily designed for per-modality recovery, they are less well suited to modality-level missingness, where imputation must rely on cross-modal evidence rather than within-modality continuity.

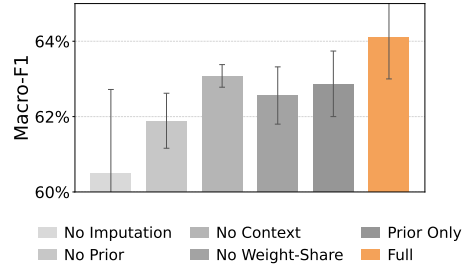


Figure 3: Strategy ablation on PTB-XL under mixed 20% missingness. All variants underperform the full model.

5 Conclusion

Missing data remains a major obstacle in multimodal time series analysis, because both within-modality missing and modality-level missing can severely degrade downstream performance. PAMF addresses this gap through prior-aware initialization in a flow-matching framework and through weight sharing with imputed-data fine-tuning that connects imputation to classification. Experiments on four datasets show that this combination delivers the strongest overall downstream performance against missing-aware baselines and direct imputation pipelines, supporting the value of distinguishing missingness structure and coupling signal recovery with downstream learning.

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A Limitations

Although PAMF improves both imputation and downstream prediction under diverse missingness settings, it still has several limitations. First, because the imputation branch is learned rather than analytically specified, the method still requires partially observed multimodal training data rich enough to learn reliable temporal and cross-modal structure. Second, our empirical study covers four public healthcare benchmarks under controlled missingness protocols, so the observed performance may depend on the benchmark preprocessing, the chosen missingness simulation, and the strength of cross-modal correlations available in each dataset.

B Broader Impacts

This work studies missing-aware multimodal time-series modeling for healthcare, so its main potential positive impact is improved robustness when some sensors or clinical measurements are unavailable. In settings where full multimodal acquisition is difficult, a model that better handles both within-modality missing and modality-level missing may help maintain downstream predictive performance and make multimodal decision support more usable in practice.

At the same time, the method also carries several risks. First, explicit imputation can produce plausible but incorrect physiological signals, which may encourage over-trust if the completed inputs are treated as ground truth rather than model estimates. Second, because the approach depends on learned temporal and cross-modal correlations, performance may degrade under distribution shift, different missingness mechanisms, or populations that are underrepresented in the training data, which could amplify unfair error patterns in deployment. Third, the target application domain involves sensitive health data, so any real-world use would require privacy-preserving data governance, secure access control, and clinical oversight. For these reasons, we view PAMF as a research-stage method rather than a deployment-ready clinical system, and any practical use should include site-specific validation and monitoring under missingness.

C Prior-Aware Source Initialization

This appendix provides the complete initialization formulas for the prior-aware source initialization x_0 introduced in Section 3.3, along with the design-choice ablation that justifies the configuration used in our main experiments.

C.1 Within-Modality Prior

For each missing position $(t, c) \in \Omega_w$, we initialize $x_0^{\text{wm}}[t, c]$ as a convex combination of the nearest observed temporal neighbor and the per-channel batch mean:

$$x_0^{\text{wm}}[t, c] = \alpha_{t,c} x_{t,c}^{\text{nbr}} + (1 - \alpha_{t,c}) \bar{x}_c^{\text{batch}}, \quad (10)$$

where $x_{t,c}^{\text{nbr}}$ is the value of the nearest observed entry in the same channel:

$$x_{t,c}^{\text{nbr}} = x_{t^*,c}, \quad t^* = \arg \min_{t': R_{t',c}^{(m)}=1} |t' - t|, \quad (11)$$

Here $R^{(m)}$ is the observation mask from Section 3.1, and $\bar{x}_c^{\text{batch}}$ is the fallback batch mean when no neighbor exists in channel c .

The blending weight $\alpha_{t,c}$ encodes confidence in the neighbor: it should be high when the gap is short and decay as the gap widens. We consider two instantiations:

Rule-based (deterministic).

$$\alpha_{t,c} = \exp(-\lambda \cdot d_{t,c}), \quad d_{t,c} = |t^* - t|, \quad (12)$$

where $\lambda > 0$ is a fixed decay coefficient (default $\lambda = 0.1$; ablated in Table 7). When no observed neighbor exists, $\alpha_{t,c} = 0$ (pure batch mean).

Learned (adaptive).

$$\alpha_{t,c} = \sigma(\text{MLP}(d_{t,c}, \mathbf{c}_{\text{imp}})), \quad (13)$$

where σ is the sigmoid function and $\mathbf{c}_{\text{imp}}$ provides cross-modal context, allowing the network to adapt the confidence based on the broader signal structure.

The rule-based variant requires no additional parameters and is theoretically grounded: as noted by Kollovich et al. [17], temporal neighbor interpolation corresponds to the posterior mean of a Gaussian process with a locally stationary kernel in the zero-noise limit, making it a principled approximation that avoids the $O(T^3)$ cost of full GP inference.

C.2 Modality-Level Prior

For modality-level missing positions $(t, c) \in \Omega_m$, no within-modality signal exists. We derive x_0^{ml} from the cross-modal context $\mathbf{c}_{\text{imp}}$ produced by h_{imp} . We consider three strategies of increasing complexity:

Batch mean (B1).

$$x_0^{\text{ml}} = \bar{x}^{\text{batch}}, \quad (14)$$

where $\bar{x}^{\text{batch}} \in \mathbb{R}^{T \times C_m}$ is the per-channel mean over all samples in the current batch that have modality m observed. This strategy is simple and parameter-free but ignores sample-level heterogeneity.

Batch matching (B2). For a query sample b with missing modality m , we retrieve the most context-similar sample in the batch that has modality m observed:

$$b^* = \arg \max_{b' \in \mathcal{B}, m \in \text{obs}(b')} \cos(\mathbf{c}_{\text{imp}}^{(b)}, \mathbf{c}_{\text{imp}}^{(b')}) , \quad x_0^{\text{ml}} = x_m^{(b^*)}, \quad (15)$$

Here $\mathcal{B}$ is the current batch, $\text{obs}(b')$ denotes the observed modalities of sample b' , and $x_m^{(b^*)}$ is modality m from the matched sample b^* . The most context-similar sample that has modality m present provides the initialization; if no such sample exists in the batch, the method falls back to B1.

Memory bank (B3). A set of P learnable prototype vectors $\mathcal{C}_m = \{c_p\}_{p=1}^P \subset \mathbb{R}^{T \times C_m}$ is maintained per modality. The initialization is a soft retrieval:

$$x_0^{\text{ml}} = \sum_{p=1}^P \pi_p c_p, \quad \pi_p = \frac{\exp(\cos(\mathbf{c}_{\text{imp}}, \tilde{c}_p))}{\sum_{p'} \exp(\cos(\mathbf{c}_{\text{imp}}, \tilde{c}_{p'}))}, \quad (16)$$

where $\tilde{c}_p$ is the key associated with prototype c_p , and π_p is its softmax retrieval weight. This variant generalizes across batches but introduces P additional parameters per modality.

These three modality-level priors reflect a clear trade-off between simplicity, sample adaptivity, and parameterization. The batch-mean baseline is maximally stable and easy to compute, but it

ignores instance-specific variation. Batch matching injects sample-specific information through cross-modal similarity, yet its quality depends on whether the current mini-batch contains a suitable reference example. The memory-bank design tries to remove that batch dependence by storing reusable prototypes, but doing so also compresses the space of possible initializations into a small learned dictionary. To make these trade-offs explicit, we report a quantitative comparison of the prior variants later in Appendix H.

D Dataset Details

D.1 Dataset Overview

The four benchmarks used in Section 4.1 cover sleep staging, ECG diagnosis, wearable activity recognition, and rhythm classification, for a total of 108,098 processed samples. Although all four datasets are physiological time series, they stress the benchmark in different ways: modality counts range from 3 grouped lead streams to 12 single-lead streams, channel counts range from 5 to 12, and sequence lengths range from 250 to 1000 time steps after preprocessing. This diversity is useful for our missingness study because it tests whether the same prior-aware and missingness-aware design can remain effective across both compact and fine-grained multimodal layouts.

To keep the appendix readable, each dataset summary below follows the same structure: source and task, modality construction, and preprocessing with split policy. This standardization makes it easier to compare benchmark assumptions across datasets without overloading the main paper’s experiment section. At the same time, the four datasets differ not only in scale but also in how the multimodal structure is defined. Sleep-EDF and PPG-DaLiA combine signals from different physiological sources, whereas PTB-XL and Chapman-Shaoxing use ECG-based modality decompositions that preserve clinically meaningful lead organization. As a result, the benchmark covers both settings where modalities are semantically distinct by nature and settings where the multimodal view is constructed from a larger sensing system to expose structured missingness.

Another important source of variation is temporal resolution. Sleep-EDF starts from long 30-second physiological epochs, PTB-XL and Chapman-Shaoxing come from fixed-length diagnostic ECG recordings, and PPG-DaLiA is built from shorter sliding windows over free-living wearable streams. These differences matter for our study because within-modality missingness and modality-level missingness can interact differently with long sequences, compact diagnostic snapshots, and motion-heavy activity windows.

D.2 Per-Dataset Details

Sleep-EDF dataset. The Sleep-EDF dataset provides overnight polysomnographic recordings with manual hypnograms and standard sleep-related physiological signals [16]. In our benchmark, it serves as a heterogeneous-but-aligned multimodal sleep-staging task, where all modalities share the same 30-second epoch but contribute different physiological evidence.

- **Task.** Five-class sleep-stage classification.
- **Labels.** ‘W’ = wakefulness; ‘N1’ = light sleep onset; ‘N2’ = stable light sleep with spindles/K-complexes; ‘N3’ = deep slow-wave sleep (with S3/S4 merged); ‘REM’ = rapid-eye-movement sleep.
- **Modalities.** We retain five channels and group them into four modalities: EEG (2 channels), EOG (1 channel), Resp (1 channel), and EMG (1 channel). This preserves clinically meaningful separation between neural, ocular, respiratory, and muscle-tone signals. Body-temperature channels available in some releases are not used.
- **Preprocessing and splits.** The loader uses fixed 30-second epochs from the benchmark-provided `split_v2.json` partition, yielding 10,918 processed samples. Each epoch is downsampled to 250 time steps and normalized with per-sample z-score scaling at load time, producing inputs of shape $\mathbb{R}^{5 \times 250}$.

PTB-XL dataset. PTB-XL is a large public 12-lead ECG benchmark with coarse diagnostic annotations over standard clinical superclasses [42]. In our benchmark, it represents a structured

ECG setting where the multimodal view is constructed from clinically related lead groups rather than distinct sensor families.

- **Task.** Multi-label ECG diagnosis prediction with five diagnostic superclasses.
- **Labels.** ‘NORM’ = normal ECG; ‘MI’ = myocardial infarction; ‘STTC’ = ST/T-wave change; ‘CD’ = conduction disturbance; ‘HYP’ = hypertrophy. A record may activate more than one label.
- **Modalities.** We split the twelve leads into three lead-group modalities: Limb (I/II/III), Augmented (aVR/aVL/aVF), and Precordial (V1–V6). This grouping gives modality-level missingness a concrete interpretation as the loss of an anatomically related lead group rather than a single isolated channel.
- **Preprocessing and splits.** The repository loader uses split files keyed by `ecg_id`. After preprocessing, each sample is represented with 12 channels and 250 time steps, i.e., $\mathbb{R}^{12 \times 250}$. Per-lead normalization is applied before classification and missingness evaluation.

PPG-DaLiA dataset. PPG-DaLiA is a wearable sensing dataset collected from 15 subjects in free-living activity settings, with synchronized chest and wrist recordings and ECG-based heart-rate reference signals [32]. In our benchmark, we use it as a multimodal wearable activity-recognition task, which makes it a natural testbed for missingness under motion-heavy, sensor-diverse conditions.

- **Task.** Nine-class activity classification.
- **Labels.** ‘transient’ = transition windows between stable activities; ‘sitting’ = seated rest; ‘ascending_stairs’ = walking upward on stairs; ‘descending_stairs’ = walking downward on stairs; ‘table_soccer’ = playing table soccer; ‘cycling’ = cycling activity; ‘driving_car’ = driving; ‘lunch_break’ = eating or quiet break period; ‘walking’ = ordinary walking.
- **Modalities.** We use five sensor modalities defined by the current code path: chest ACC (3 axes), wrist ACC (3 axes), wrist BVP, wrist EDA, and wrist TEMP. This gives 9 total channels and preserves the distinction between motion-dominant and physiology-dominant wearable measurements rather than merging them into a single generic stream.
- **Preprocessing and splits.** The benchmark loader operates on segmented 8-second windows mapped to a common 256-step grid, uses benchmark split JSONs, and produces 64,697 processed segments in total. Per-channel z-score normalization is applied within each segment, giving final inputs of shape $\mathbb{R}^{9 \times 256}$.

Chapman-Shaoxing dataset. Chapman-Shaoxing is a 12-lead ECG corpus with diagnostic annotations covering common and additional cardiovascular conditions [54]. In our benchmark-aligned configuration, it complements PTB-XL by exposing a more fine-grained lead-wise multimodal formulation, where each ECG lead is treated as its own modality.

- **Task.** Multi-label diagnosis prediction with seven superclasses.
- **Labels.** ‘normal’ = normal rhythm or morphology; ‘cd’ = conduction disturbance; ‘mi’ = myocardial infarction; ‘sttc’ = ST/T-wave change; ‘other’ = other cardiac abnormality not covered by the previous groups; ‘afib’ = atrial fibrillation; ‘hyp’ = hypertrophy. As in PTB-XL, a record may activate multiple labels.
- **Modalities.** We treat each lead as its own modality: I, II, III, aVR, aVL, aVF, and V1–V6. This yields 12 single-lead modalities, so modality-level missingness corresponds to the loss of an individual lead rather than a coarse group.
- **Preprocessing and splits.** The loader resamples each 10-second record from 500 Hz to 100 Hz, keeps 1000 time steps per lead, applies per-channel z-score normalization, and reads the benchmark `climb_splits` for train/val/test evaluation. If those split files are regenerated from the CSV release, the code maps the CSV `val` partition to test and repartitions the CSV `train` subset into train/val with a 90/10 split and seed 42. The resulting tensor shape is $\mathbb{R}^{12 \times 1000}$.

Table 5: Summary of the four benchmark datasets used in this work. Sample counts and tensor shapes correspond to the processed benchmark inputs used by our loaders rather than the raw source files.

Dataset	Domain	Task	Samples	Modalities	Channels	Processed shape
Sleep-EDF	Sleep staging	5-class classification	10,918	4	5	$\mathbb{R}^{5 \times 250}$
PTB-XL	ECG diagnosis	5-label prediction	21,837	3	12	$\mathbb{R}^{12 \times 250}$
PPG-DaLiA	Wearable activity recognition	9-class classification	64,697	5	9	$\mathbb{R}^{9 \times 256}$
Chapman-Shaoxing	ECG rhythm/diagnosis	7-label prediction	10,646	12	12	$\mathbb{R}^{12 \times 1000}$
Total / Range	4 tasks	—	108,098	3–12	5–12	$T = 250\text{--}1000$

D.3 Overall Summary Table

Table 5 consolidates the benchmark properties used throughout our experiments. It is intended as a quick reference for dataset scale, modality design, and processed input size, complementing the longer descriptions above. In particular, it highlights that the benchmark spans both coarse and fine-grained modality decompositions, moderate and large sample counts, and short to relatively long processed sequences, which together define the scope of the missingness scenarios studied in the paper.

D.4 Missingness Protocol Details

The main paper reports within-modality missing, modality-level missing, mixed missingness, and high-missingness stress tests. Here we summarize how those conditions are implemented in the benchmark pipeline. Missingness is injected at data-loading time rather than written out as a separate copy of every dataset, so different missingness settings can be evaluated without duplicating the underlying processed signals on disk. To keep the stochastic patterns reproducible, the injector constructs a local random generator from the pair (`seed`, `sample_id`), ensuring that the same sample receives the same missingness pattern whenever the same configuration is rerun.

Modality-level missingness. For modality-level missingness, each currently present modality group is dropped independently with Bernoulli probability p_{mod} . In the main experiments, we set $p_{\text{mod}} = 0.2$ for the basic modality-level missing setting. A safeguard enforces that at least one modality remains: if all groups would be dropped, one dropped group is restored uniformly at random. Once a group is removed, its entry in `mod_mask` is set to zero and the corresponding row block in the temporal mask is zeroed as well. The meaning of one dropped modality therefore depends on the benchmark-specific grouping: one of four physiological groups in Sleep-EDF, one of three lead groups in PTB-XL, one of five wearable sensor streams in PPG-DaLiA, or one of twelve single leads in Chapman-Shaoxing.

Within-modality missingness. For within-modality missingness, we use contiguous time-block masking rather than scattered point deletion. For a sequence of length T , each block length is sampled from the interval $[0.05T, 0.10T]$, matching the 5–10% range described in Section 4.1. The number of blocks is chosen to reach approximately 20% total temporal corruption, and non-overlap is enforced by rejection sampling with a bounded retry budget. These blocks are applied independently across time-series rows, so different channels or modality groups typically lose different temporal windows. This design better mimics sensor dropouts and short recording interruptions than iid masking would.

Mixed and high missingness settings. The mixed setting combines the same two mechanisms: 20% block-based within-modality missing together with 20% Bernoulli modality-level missing. In implementation, the temporal block mask is constructed first and then any fully dropped modalities overwrite their corresponding temporal rows, so modality removal takes precedence whenever both types affect the same group. For the high-missingness stress test, we increase the corruption rate to 50% while keeping the same missingness semantics, allowing us to measure whether performance degrades gracefully when observation loss becomes severe.

E Evaluation Metric Details

For all classification experiments, we report macro-averaged F1 and macro-averaged AUROC. Let N be the number of test samples and K the number of classes. For class k , let TP_k , FP_k , and FN_k denote the numbers of true positives, false positives, and false negatives, respectively. The class-wise precision and recall are

$$P_k = \frac{TP_k}{TP_k + FP_k}, \quad R_k = \frac{TP_k}{TP_k + FN_k}, \quad (17)$$

and the class-wise F1 score is

$$F1_k = \frac{2P_k R_k}{P_k + R_k}. \quad (18)$$

The reported macro-F1 is the unweighted average over classes,

$$\text{Macro-F1} = \frac{1}{K} \sum_{k=1}^K F1_k. \quad (19)$$

For AUROC, we use a one-vs.-rest formulation and then average across classes. Let $y_{i,k} \in \{0, 1\}$ denote the ground-truth indicator of whether sample i belongs to class k , and let $s_{i,k}$ be the predicted confidence score for that class. The class-wise AUROC is

$$\text{AUROC}_k = \frac{1}{|\mathcal{P}_k| |\mathcal{N}_k|} \sum_{i \in \mathcal{P}_k} \sum_{j \in \mathcal{N}_k} \mathbb{I}(s_{i,k} > s_{j,k}), \quad (20)$$

where $\mathcal{P}_k = \{i \mid y_{i,k} = 1\}$ and $\mathcal{N}_k = \{i \mid y_{i,k} = 0\}$ are the positive and negative sample sets for class k . The reported macro-AUROC is

$$\text{Macro-AUROC} = \frac{1}{K} \sum_{k=1}^K \text{AUROC}_k. \quad (21)$$

This matches our implementation, which computes macro-F1 and macro AUROC with a one-vs.-rest reduction across classes.

F Baseline Methods

This section discusses how the representative baselines used in our experiments handle missing inputs and, more importantly, what they still leave unresolved. The core distinction in our paper is between methods that tolerate missingness at the representation level and methods that explicitly reconstruct missing signal content before downstream prediction.

Flex-MoE [51]. Flex-MoE is one of the clearest examples of a missing-aware fusion model. It explicitly organizes computation around modality subsets through its routers and missing-modality bank, so it is built for the combinatorial setting in which different samples arrive with different observed-modality patterns. Among the classification baselines, this makes it particularly relevant for modality-level missing, because the model does not assume that all modalities are present at test time. Even so, the substitute for a missing modality remains a learned subset-aware representation rather than a sample-level reconstruction of the missing signal itself.

- **What it solves.** It handles arbitrary observed-modality subsets explicitly and learns routing behavior that remains effective when whole modalities are absent.
- **What it does not solve.** It does not reconstruct the missing signal trajectory; the missing-modality bank provides a learned placeholder for fusion, not an imputed waveform or sensor trace tied to the current sample.
- **Why this matters here.** This makes Flex-MoE a strong baseline for missing-aware classification, but not a direct answer to the question of whether explicit signal recovery can provide more downstream value.

MIRA [18]. MIRA is stronger than ordinary fusion baselines in the sense that it natively handles irregular sampling, sparse observations, and masked values within a large time-series foundation-model framework. Its continuous-time positional design and expert specialization make it well matched to temporally sparse clinical signals, and in our benchmark it is the baseline that comes closest to treating missingness as part of the time-series learning problem rather than as an afterthought. Still, MIRA remains prediction-oriented: it uses masks to tell the encoder which observations are reliable, but it does not build an explicit sample-specific recovery process for unobserved signals.

- **What it solves.** It can process partially observed time series without heavy dataset-specific redesign and is naturally robust to irregular temporal sparsity.
- **What it does not solve.** It does not explicitly reconstruct missing signal values, and it does not separate within-modality missing from modality-level missing through different source priors or recovery paths.
- **Why this matters here.** Our setting distinguishes between local gaps inside an observed modality and the complete absence of a modality, whereas MIRA mainly treats both through masked observation encoding.

FuseMoE [11]. FuseMoE is a cross-modal Transformer with sparse expert routing, so it is designed to keep multimodal fusion effective even when different streams have different reliability or sampling structure. In our setting, its main advantage is that missing modalities can be represented by masked or placeholder tokens while the observed modalities continue to interact through cross-attention. This makes it a strong benchmark for missing-aware fusion under modality-level missing, especially when the remaining modalities still contain enough evidence for classification.

- **What it solves.** It learns expert-specialized fusion over the observed subset and can continue prediction without requiring every modality to be present.
- **What it does not solve.** The missing modality is bypassed rather than reconstructed, so the model never produces a sample-specific estimate of the absent stream.
- **Why this matters here.** When the missing modality contains complementary rather than redundant information, missing-aware routing alone may preserve stability but still leave task-relevant evidence unrecovered.

Maestro [28]. Maestro is the most explicitly robustness-oriented classifier among the fusion baselines because it combines sparse attention, modality-aware tokenization, and curriculum modality dropout. That training strategy exposes the model to progressively harder missingness patterns and helps it learn to classify from partial evidence instead of collapsing when an input stream disappears. In our benchmark, this makes Maestro a particularly strong representative of the “tolerate missingness” family.

- **What it solves.** It improves downstream robustness to partial observation through training-time modality dropout and missing-aware token handling.
- **What it does not solve.** Its robustness comes from better classification under absent inputs, not from reconstructing the missing waveform or sensor content itself.
- **Why this matters here.** Maestro tests how far carefully trained missing-aware fusion can go, but it still leaves open whether explicit recovery can add information beyond robust masking-based classification.

CSDI [39]. CSDI addresses the opposite side of the problem from the fusion baselines above: it is an explicit imputer. Its role in our benchmark is to provide a principled diffusion-based reconstruction baseline that fills in missing values before the downstream classifier is applied. This makes it highly relevant for within-modality missing, where the central question is not just whether the classifier can ignore corrupted segments, but whether the underlying signal can be recovered well enough to improve the final prediction.

- **What it solves.** It explicitly reconstructs missing values and therefore targets signal recovery directly rather than only masking or bypassing corrupted inputs.

- **What it does not solve.** It is primarily designed for within-series imputation and does not directly address multimodal modality-level absence with different priors for different missingness mechanisms.
- **Why this matters here.** CSDI is a strong test of whether generic diffusion-based imputation helps, but not of whether missingness-type-aware initialization and downstream-coupled training help further.

SSSD [26]. SSSD is also an explicit diffusion-style imputer, but with a structured state-space design intended to model temporal dependencies more efficiently. Within our benchmark, it serves as a second reconstruction baseline that tests whether stronger temporal generative modeling alone is sufficient to close the gap under missingness. Like CSDI, however, its focus is still generic signal completion rather than multimodal missingness reasoning.

- **What it solves.** It reconstructs missing signal values with a dedicated generative model and therefore provides a direct signal-recovery comparison to our method.
- **What it does not solve.** It does not naturally distinguish within-modality missing from modality-level missing, and its imputation stage remains separate from downstream task supervision.
- **Why this matters here.** SSSD helps test whether better generic imputation is enough, but it does not answer whether recovery should be conditioned on missingness type and tied more tightly to the end task.

Summary of the gap. Taken together, the baselines in our study split into two incomplete families. Missing-aware fusion models can classify with absent inputs but usually do not reconstruct the missing signal, whereas explicit imputers can reconstruct signals but are not built to handle multimodal modality-level absence with type-specific priors or downstream-coupled training. This is precisely the gap that motivates PAMF in the main paper. The benchmark is therefore not just comparing stronger versus weaker classifiers; it is comparing three different responses to missing data: tolerating it, generically imputing it, or explicitly reconstructing it with priors matched to the missingness type while still optimizing for the downstream task.

G Implementation Details

Module roles. The encoder, f_{fm} , and f_{ds} serve different purposes in PAMF. The shared encoder backbone is instantiated twice, as h_{imp} for cross-modal conditioning and as h_{fus} for downstream prediction. In both cases, each modality is linearly projected, enriched with temporal positional encoding, processed by two per-modality Informer layers, and then fused across modalities by a cross-modal MoE block with Informer attention [55]. The resulting fused vector is used either as the imputation context $\mathbf{c}_{\text{imp}}$ or as the downstream-task context $\mathbf{c}_{\text{fus}}$.

f_{fm} is implemented as a conditional velocity network. Given the interpolated state x_τ , it combines a continuous FM time embedding with side information built from time embeddings, feature embeddings, the observation mask, and $\mathbf{c}_{\text{imp}}$, then predicts the velocity through four residual blocks with alternating time- and feature-wise attention. During training, we sample one continuous FM time $\tau \sim \mathcal{U}[0, 1]$ independently for each sample in the batch, construct $x_\tau = (1 - \tau)x_0 + \tau x_1$, and inject τ into the network through a sinusoidal continuous-time embedding followed by a two-layer MLP. At inference time, we discretize $[0, 1]$ into 20 uniform Euler steps and evaluate the velocity field on that fixed grid. Finally, f_{ds} is a lightweight residual MLP head consisting of two linear projections with ReLU and dropout, followed by a linear output layer that maps $\mathbf{c}_{\text{fus}}$ to the task label.

Hyperparameter. Table 6 summarizes the shared model and training configurations used in all main experiments, including the common three-phase schedule of 5/20/20 epochs.

Checkpoint selection. For the standard three-phase curriculum, Phase 2 saves `best_phase2_diffusion.pt` and Phase 3 is initialized from that best Phase 2 checkpoint rather than from the final epoch indiscriminately. Phase 3 then saves `best_phase3_joint.pt`, which is the checkpoint used for all final test-set evaluation scripts. The selection criteria differ by stage: the diffusion phase tracks the best validation RMSE because its goal is signal-level recovery,

Table 6: Model and training configurations used for PAMF. The encoder output dimension fused_dim equals $N_{\text{mod}} \times 128$.

Configuration Type	Parameter Name / Value
Encoder	Per-modal projection dimension: 128 Per-modal Informer layers: 2 Per-modal attention heads: 2 Cross-modal MoE blocks with Informer attention: 1 Cross-modal MoE experts: 4 Cross-modal expert hidden dimension: 64 Encoder output dimension (fused_dim): $N_{\text{mod}} \times 128$ Classifier dropout: 0.1
Flow Matching	Residual channels: 64 Number of residual blocks: 4 FM time embedding dimension: 128 Feature embedding dimension: 64 Time-attention heads: 8 Side-information dimension: $128 + 64 + 1 + \text{fused_dim}$ Euler ODE steps: 20
Training and Prior	Phase 1 epochs: 5 Phase 2 epochs: 20 Phase 3 epochs: 20 Batch size: 64 Classification learning rate: 1e-3 FM learning rate: 5e-5 Prior initialization: neighbor_rule + batch_mean Phase 2 checkpoint selection: lowest validation RMSE Phase 3 checkpoint selection: highest validation Macro-F1 Random seeds: 42 / 52 / 62

whereas the final classification phase tracks the best validation Macro-F1 on the downstream task. This two-stage selection rule keeps the hand-off between imputation and classification aligned with the objective of each phase instead of forcing one metric to govern the entire curriculum.

Training hardware. All experiments were trained on NVIDIA DGX B200 systems, using one NVIDIA B200 GPU per run. For the common three-phase schedule of 5/20/20 epochs, the representative wall-clock training time was approximately 55.2 minutes on PPG-DaLiA (median over three runs), 9.8 hours on Sleep-EDF, 25.3 minutes on PTB-XL, and 1.64 hours on Chapman-Shaoxing.

H Additional Results

This section collects appendix-only analyses that elaborate on design choices supporting the main paper while remaining secondary to the core narrative. We focus on secondary quantitative results that help interpret design choices and complete the experimental picture, but that are not essential for understanding the main method or its headline performance. We begin with a direct ablation of the prior initialization strategies introduced in Section C.

H.1 Design Choice Ablation

Table 7 reports the effect of prior initialization choices on downstream classification under the corresponding missingness type. The goal of this ablation is not to maximize every appendix number with a more complicated prior, but to test whether additional adaptivity consistently translates into better downstream behavior. Because the proposed framework uses the prior only as a source initialization for flow matching, a stronger prior should ideally shorten the transport path without introducing instability or overfitting. The table therefore helps distinguish improvements that are structurally meaningful from those that merely add parameters.

Table 7: Ablation of prior initialization strategies on PTB-XL. within-modality missing variants compare neighbor-rule initialization and learned-neighbor initialization under within-modality missing 20%. modality-level missing variants compare batch-matched, batch-mean (default), and memory-bank initialization under modality-level missing 20%. Means and standard deviations over three runs are rounded to three decimals. Best per group in **bold**.

Missing type	Prior variant	Macro-F1
within-modality missing	Neighbor rule	0.661 $\pm$ 0.003
	Neighbor learned	0.668 $\pm$ 0.007
modality-level missing	Batch-matched	0.650 $\pm$ 0.002
	Batch-mean (default)	0.648 $\pm$ 0.002
	Memory Bank	0.475 $\pm$ 0.002

For the refreshed PTB-XL modality-level missing reruns at $d_{\text{modal}} = 128$, the corresponding AUROC values are 0.880 $\pm$ 0.001 for batch-matched and 0.878 $\pm$ 0.002 for the default batch-mean prior. In contrast, the learned memory-bank prior performs much worse, likely because a small set of learned prototypes collapses toward overly averaged modality-level templates and loses the sample-specific cross-modal structure needed for reliable initialization.

The within-modality missing results suggest that adding limited adaptivity can be helpful when the relevant uncertainty is still strongly local: the learned-neighbor variant slightly outperforms the rule-based decay, which is consistent with the idea that the model can refine how much trust to place in nearby observations without abandoning the temporal-neighbor inductive bias. For modality-level missing, the picture is different. Batch-matched retrieval performs only marginally better than the parameter-free batch mean, indicating that once an entire modality is absent, coarse cross-sample context is useful but does not dramatically change the downstream outcome. The large drop of the memory-bank variant further suggests that introducing learnable global prototypes is not automatically beneficial: if those prototypes fail to preserve sample-specific cross-modal alignment, they can become worse initial anchors than even a simple global average.

Default configuration. In the main experiments, we use the rule-based within-modality prior and the batch-mean modality-level prior. Appendix H.1 shows that batch-matched retrieval is only slightly stronger than the default batch mean, whereas the learned memory-bank variant performs substantially worse. We therefore keep the parameter-free batch-mean prior in the primary setting because it remains competitive without introducing additional learned prototypes. More broadly, these results support our main design principle that prior-aware initialization should encode the right missingness structure with as little unnecessary complexity as possible.

H.2 Exact HPO Values Used in Figure 2

Table 8 lists the paper-facing mean and standard deviation values used to generate Figure 2. We report these values directly from the plotting source so that the appendix matches the figure exactly, even though the broader HPO archive in `results_overall.typ` still contains historical entries that are not fully synchronized with the final paper-facing plot.

The table makes the two trends in Figure 2 explicit. Increasing the number of experts beyond $N=4$ does not deliver a stable gain, and overly large expert dimensionality quickly becomes harmful: performance improves from $d_{\text{modal}}=16$ to 128, but collapses at 256. These exact values support the final choice of $N=4$ and $d_{\text{modal}}=128$ in the main configuration.

H.3 Clean-Input Performance

Table 9 reports clean-input performance on the three benchmarks for which we currently maintain complete clean summaries in the benchmark archive: Sleep-EDF, PTB-XL, and PPG-DaLiA. We include this table to show that the missing-aware design of PAMF does not depend on evaluating only corrupted inputs. To keep this appendix table aligned with the main-paper comparison set, we report the same primary baselines as in the main results and do not include ShaSpec or CLIMB here. Even without synthetic missingness, the model remains competitive and achieves the best Macro-F1 on

Table 8: Exact HPO values used in Figure 2 on PTB-XL under mixed 20% missingness. The exploratory $N=1$ point is omitted to match the plotted figure. Means and standard deviations are reported exactly from the paper-facing plotting source.

Sweep	Setting	Macro-F1	AUROC
<i>Number of experts N</i>			
N	2	0.618 $\pm$ 0.006	0.850 $\pm$ 0.005
N	4	0.641 $\pm$ 0.011	0.867 $\pm$ 0.000
N	8	0.622 $\pm$ 0.010	0.861 $\pm$ 0.002
N	12	0.616 $\pm$ 0.007	0.848 $\pm$ 0.002
N	16	0.605 $\pm$ 0.026	0.851 $\pm$ 0.005
N	32	0.617 $\pm$ 0.008	0.847 $\pm$ 0.004
<i>Expert size d_{modal}</i>			
d_{modal}	16	0.586 $\pm$ 0.011	0.846 $\pm$ 0.004
d_{modal}	32	0.622 $\pm$ 0.008	0.851 $\pm$ 0.002
d_{modal}	64	0.612 $\pm$ 0.022	0.842 $\pm$ 0.005
d_{modal}	128	0.641 $\pm$ 0.011	0.867 $\pm$ 0.000
d_{modal}	256	0.311 $\pm$ 0.132	0.786 $\pm$ 0.017

Table 9: Clean-input classification performance on the three benchmarks with complete clean summaries in the current archive. To match the main-paper comparison set, the table reports the same primary baselines and excludes ShaSpec and CLIMB. Means and standard deviations over three runs are rounded to three decimals. **Bold**: best; underlined: second-best within the reported comparison set.

Method	Sleep-EDF		PTB-XL		PPG-DaLiA	
	F1	AUROC	F1	AUROC	F1	AUROC
FuseMoE	0.582 $\pm$ 0.004	0.930 $\pm$ 0.001	0.224 $\pm$ 0.034	0.686 $\pm$ 0.023	0.623 $\pm$ 0.019	0.925 $\pm$ 0.002
Flex-MoE	<u>0.704</u> $\pm$ 0.011	<u>0.973</u> $\pm$ 0.002	0.440 $\pm$ 0.016	0.767 $\pm$ 0.004	0.624 $\pm$ 0.004	0.928 $\pm$ 0.001
Maestro	0.699 $\pm$ 0.004	0.971 $\pm$ 0.002	<u>0.622</u> $\pm$ 0.009	<u>0.882</u> $\pm$ 0.002	0.769 $\pm$ 0.004	0.958 $\pm$ 0.002
MIRA	0.629 $\pm$ 0.000	0.958 $\pm$ 0.001	0.482 $\pm$ 0.022	0.826 $\pm$ 0.006	0.676 $\pm$ 0.004	0.939 $\pm$ 0.002
PAMF	0.744 $\pm$ 0.007	0.981 $\pm$ 0.002	0.657 $\pm$ 0.005	0.884 $\pm$ 0.002	<u>0.748</u> $\pm$ 0.011	<u>0.951</u> $\pm$ 0.003

Sleep-EDF and PTB-XL within this primary comparison set, while remaining close to the strongest baseline on PPG-DaLiA.

These clean results provide a useful complement to the missing-input tables in the main paper. Within the primary baseline set used throughout the main text, PAMF remains the strongest method on Sleep-EDF and PTB-XL even without injected missingness, suggesting that the shared encoder and imputed-data fine-tuning strategy do not trade away clean-case classification quality. On PPG-DaLiA, Maestro remains the strongest baseline, but PAMF still ranks second on both Macro-F1 and AUROC within the same comparison set.